

WEIFANG, China

WMO: 548430

Lat: 36.767N

Lon: 119.183E

Elev: 72

StdP: 14.66

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.3	16.4	-4.8	4.3	26.6	-1.0	5.2	25.0	21.0	29.2	18.7	30.4	5.5	290	0.438

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.8	94.3	75.5	91.9	75.2	89.6	74.7	81.5	89.3	80.1	86.9	78.8	84.9	7.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.4	153.0	86.2	78.2	147.1	84.8	77.0	141.0	83.1	45.5	89.9	44.0	86.9	42.5	85.1	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.9	16.3	14.1	8.9	99.3	4.4	2.8	5.7	101.3	3.1	102.9	0.6	104.5	-2.6	106.5		
(5)			7.4	83.3	3.9	1.4	4.6	84.3	2.3	85.1	0.1	85.9	-2.7	86.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.4	29.0	34.2	44.9	56.7	67.3	75.8	80.6	78.6	70.6	59.5	45.1	33.2		
	DBStd	18.93	5.90	7.03	8.27	7.38	6.32	4.85	4.19	4.38	5.20	6.38	7.44	6.15		
	HDD50	2032	651	444	203	21	0	0	0	0	0	7	185	521		
	HDD65	4700	1116	864	624	263	46	1	0	0	13	189	598	986		
	CDD50	4372	0	0	45	223	538	773	949	887	617	303	37	0		
	CDD65	1564	0	0	1	15	119	324	484	422	180	19	0	0		
	CDH74	15009	0	0	22	214	1272	3082	5040	3957	1275	146	1	0		
	CDH80	5686	0	0	3	43	439	1326	2139	1401	320	15	0	0		
(14) Wind	WSAvg	6.2	6.0	6.5	7.4	8.0	7.1	6.9	5.9	5.0	4.9	5.3	5.9	5.8		
(15) Precipitation	PrecAvg	23.4	0.2	0.5	0.6	1.1	2.0	3.1	5.5	5.6	2.3	1.1	1.0	0.4		
	PrecMax	36.2	1.1	1.3	1.6	2.6	4.8	7.1	11.2	11.9	6.4	3.9	4.0	1.4		
	PrecMin	13.1	0.0	0.0	0.0	0.1	0.0	0.2	0.6	0.6	0.2	0.1	0.0	0.0		
	PrecStd	5.4	0.3	0.4	0.4	0.6	1.1	1.9	3.0	3.0	1.7	1.0	1.0	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.2	60.2	77.0	85.1	95.0	97.6	97.7	94.0	89.6	81.9	70.8	55.4		
		MCWB	39.5	46.1	56.2	63.8	67.2	68.8	78.1	79.7	72.3	62.9	55.6	45.4		
	2%	DB	46.1	54.6	70.4	79.7	88.6	93.6	94.7	91.4	86.1	78.0	65.6	50.6		
		MCWB	36.8	42.7	52.6	60.7	65.8	70.2	78.0	79.0	70.0	61.9	53.0	42.2		
	5%	DB	42.5	50.8	64.7	75.6	84.8	90.8	92.4	89.2	83.6	74.6	61.2	46.7		
		MCWB	34.6	40.4	49.5	58.7	64.4	69.9	78.2	77.4	69.0	60.2	51.1	39.3		
	10%	DB	39.1	46.9	59.4	71.4	81.2	87.6	89.8	87.0	81.1	71.1	57.6	43.2		
		MCWB	32.6	38.0	46.2	56.7	63.4	69.5	77.3	76.6	67.9	59.2	49.7	37.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.0	48.5	58.5	67.7	71.7	78.2	83.4	82.7	78.4	68.3	58.6	48.5		
		MCDB	47.1	56.1	72.9	79.4	85.0	87.7	92.1	90.4	84.4	74.5	64.0	52.6		
	2%	WB	38.3	44.7	54.2	63.6	69.7	75.6	81.9	81.5	74.7	65.7	55.9	43.8		
		MCDB	44.2	52.9	67.1	75.5	82.6	84.6	90.4	88.6	80.9	71.9	62.3	48.5		
	5%	WB	35.6	41.8	51.0	60.8	67.6	73.8	80.7	80.2	72.3	63.6	53.1	40.4		
		MCDB	41.0	48.5	62.1	72.2	79.6	83.2	88.3	86.3	78.9	70.3	59.1	44.9		
	10%	WB	33.1	38.9	47.8	58.1	65.6	72.1	79.3	78.9	70.7	61.4	50.4	37.8		
		MCDB	38.2	45.9	57.7	68.0	76.8	81.7	86.3	84.1	77.5	68.6	56.9	42.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.7	18.4	20.8	21.4	21.3	19.5	14.8	13.9	17.9	19.1	18.0	16.4		
		MCDBR	22.7	24.6	28.0	28.1	27.6	25.3	19.2	16.3	21.2	23.9	23.8	22.1		
		MCWBR	14.8	14.8	14.6	13.5	10.8	8.2	6.7	6.3	7.8	10.0	12.8	14.2		
	5% WB	MCDBR	19.8	21.5	25.6	24.9	23.1	18.3	15.2	13.0	16.2	17.9	19.0	18.2		
		MCWBR	13.6	13.8	14.1	13.1	10.9	7.8	6.8	6.2	7.5	9.3	12.0	13.1		
(40) Clear-Sky Solar Irradiance	taub		0.518	0.577	0.611	0.632	0.654	0.771	0.752	0.690	0.618	0.647	0.584	0.524		
	taud		1.749	1.666	1.622	1.626	1.622	1.467	1.551	1.649	1.733	1.615	1.683	1.765		
	Ebn at Noon		194	200	211	217	215	191	193	202	207	180	173	181		
	Edh at Noon		55	67	77	80	82	95	87	78	68	70	58	51		
(44) All-Sky Solar Radiation	RadAvg		752	971	1390	1626	1808	1693	1499	1444	1318	1054	790	685		
	RadStd		84	110	100	68	132	124	141	142	107	98	103	63		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.33	+2.14	N/A	N/A	N/A	N/A	-175	-290	+322	+178
(47) Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+133	N/A

Nomenclature: See separate page